



TECHNOLOGY
SOLUTIONS UK LTD
part of **HID**

USER GUIDE: 1128 *BLUETOOTH*[®] UHF RFID READER



www.tsl.com

Design • Development • Manufacture

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INTRODUCTION

The TSL 1128 *Bluetooth*® UHF RFID Reader provides Ultra High Frequency (UHF) Radio Frequency Identification (RFID), with optional barcode scanning functionality. The unit can be used in batch mode using the optional Micro SD card, or connected to a host device via *Bluetooth*®. The 1128 can read and write to EPC Global Class 1 Gen 2 UHF RFID transponders.

PARTS OF THE 1128 UHF READER

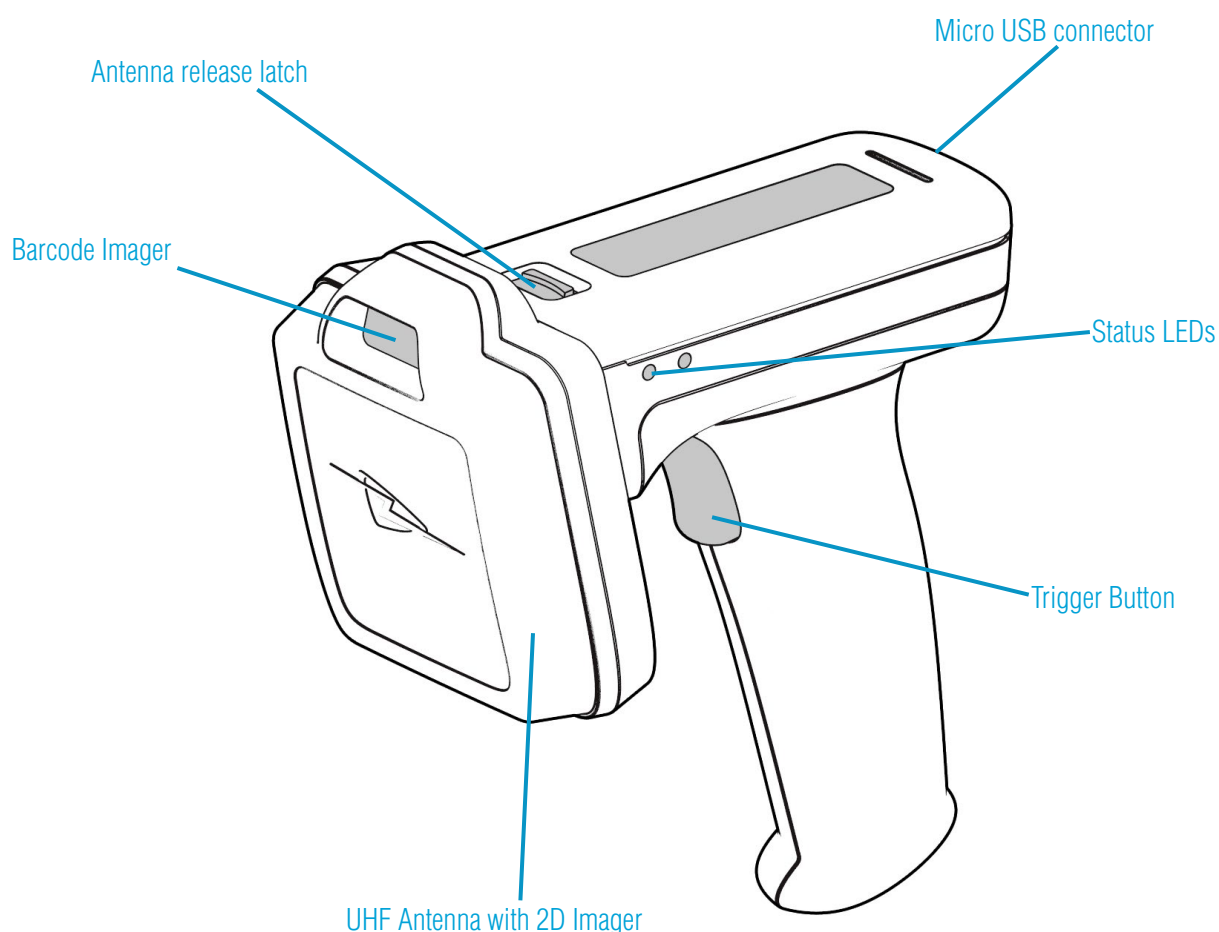


Figure 1: Parts of the 1128 UHF Reader

ALTERNATIVE CONFIGURATION OF THE 1128 UHF READER

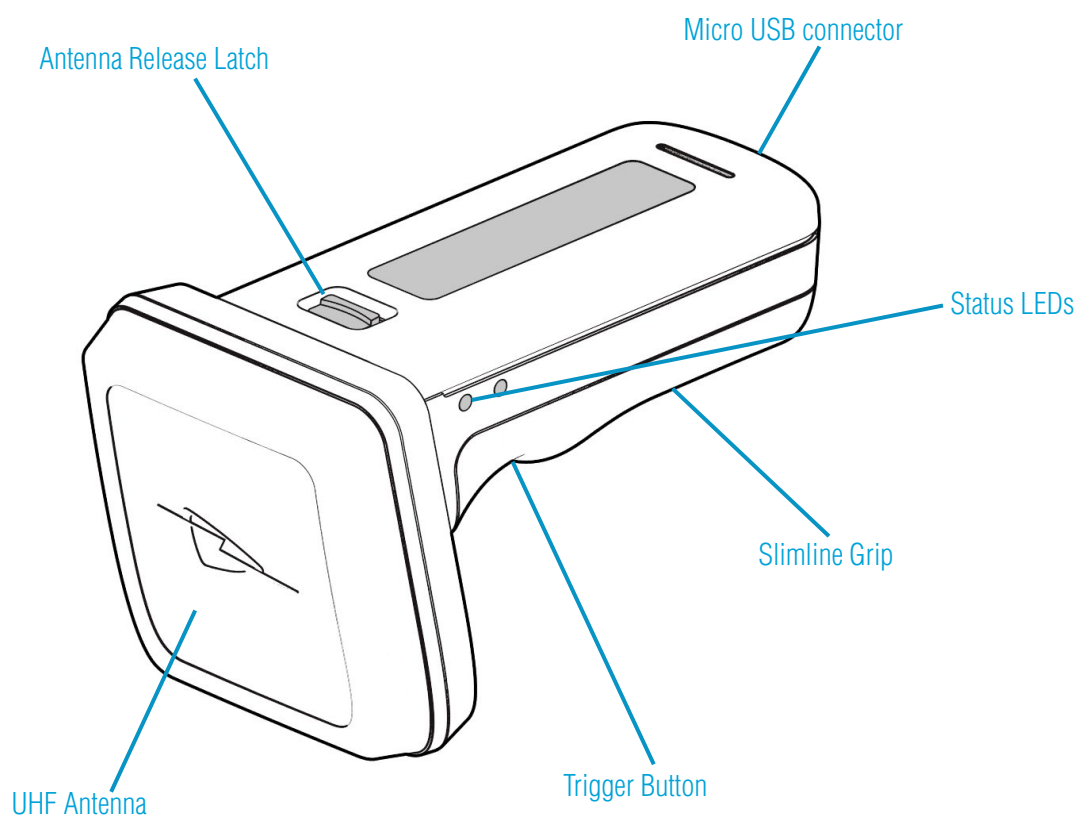


Figure 2: Parts of the 1128 UHF Reader

USING THE 1128 UHF READER

BATTERY INSTALLATION OR REMOVAL

The battery is charged using the supplied micro USB lead and therefore is unlikely to need to be changed once installed. To access the battery compartment the grip handle must first be removed.

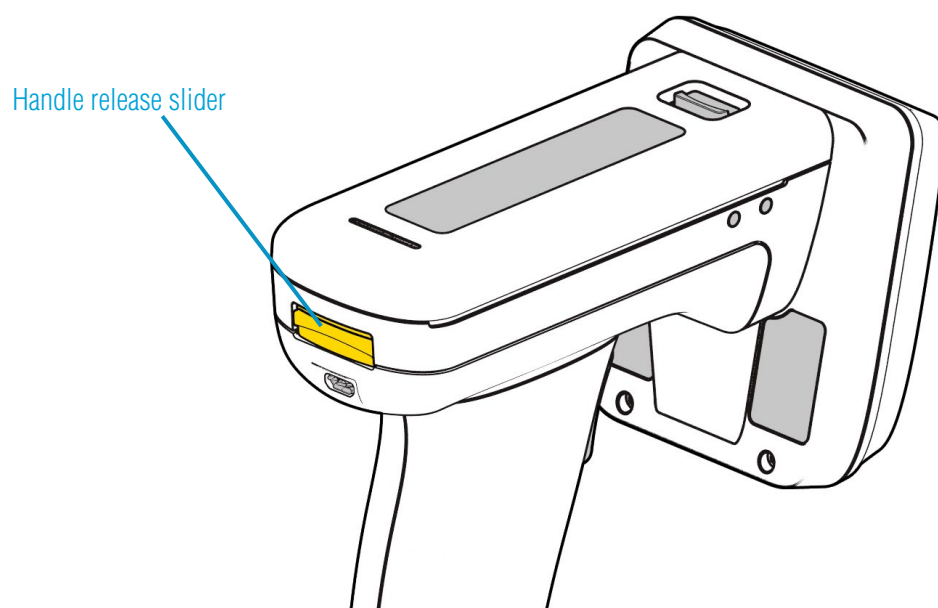


Figure 3: Remove handle

The battery compartment has a release latch. The direction to push is shown in the diagram below.

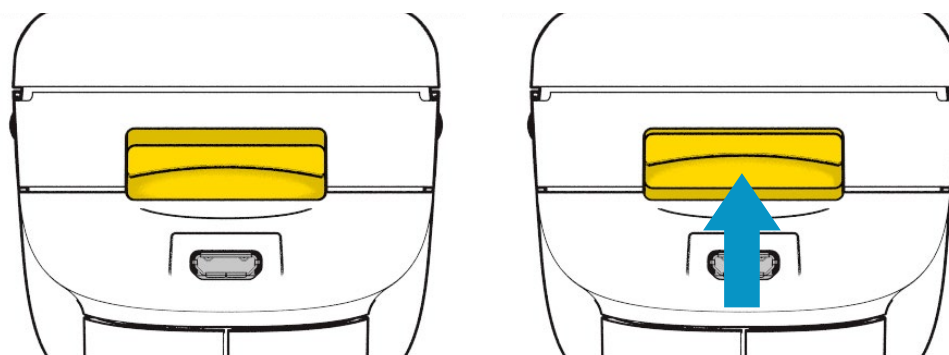


Figure 4: Handle release latch operation

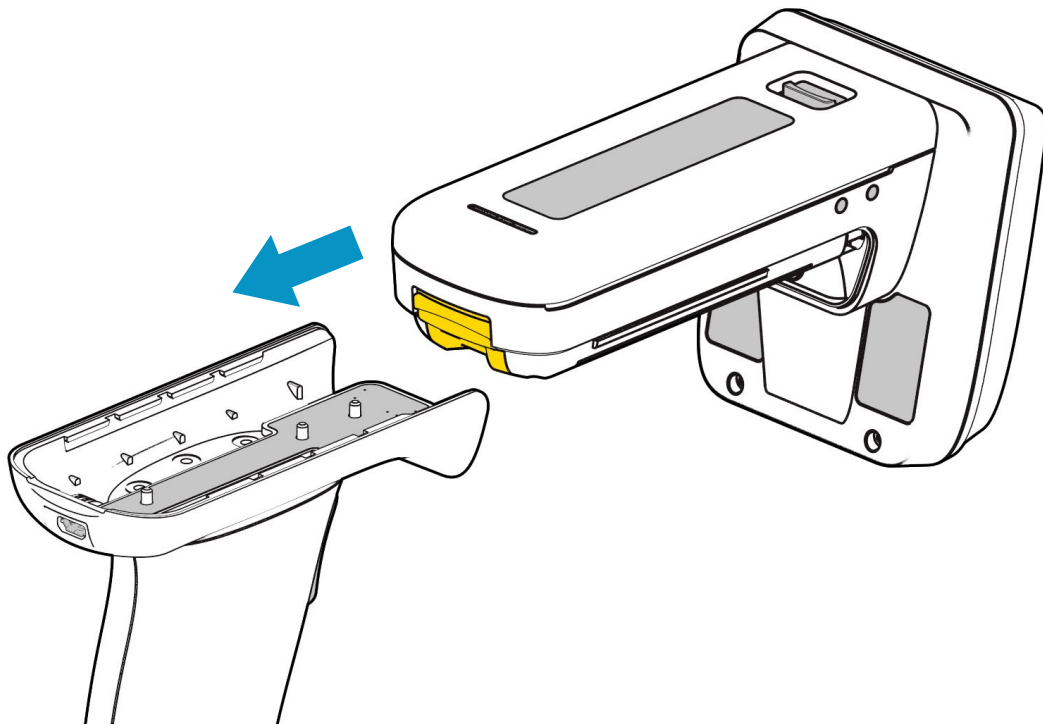


Figure 5: Slide handle off

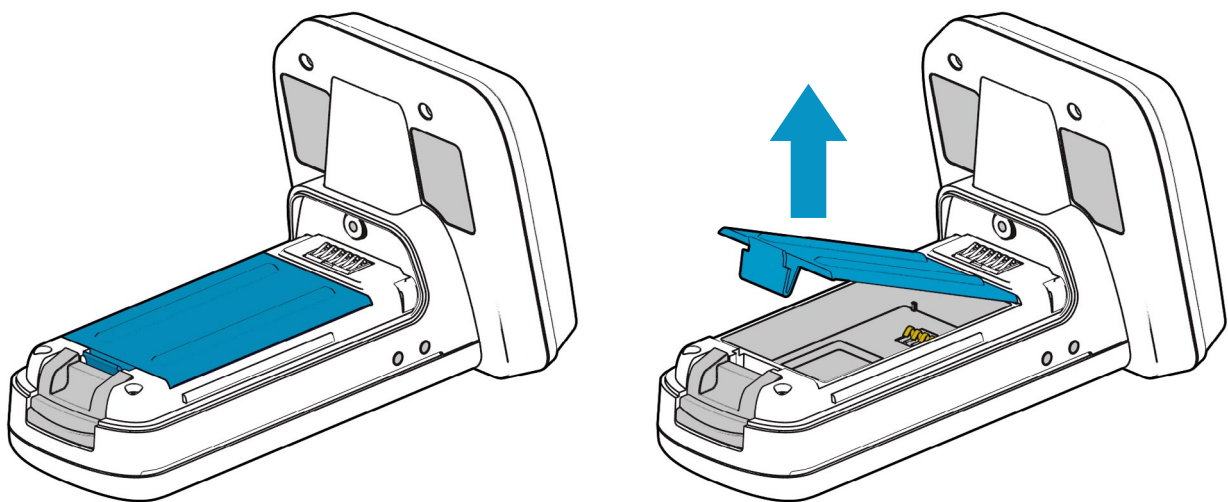


Figure 6: Remove battery cover

The battery contacts need to be face down at the top of the battery to meet the contacts shown in the image below.

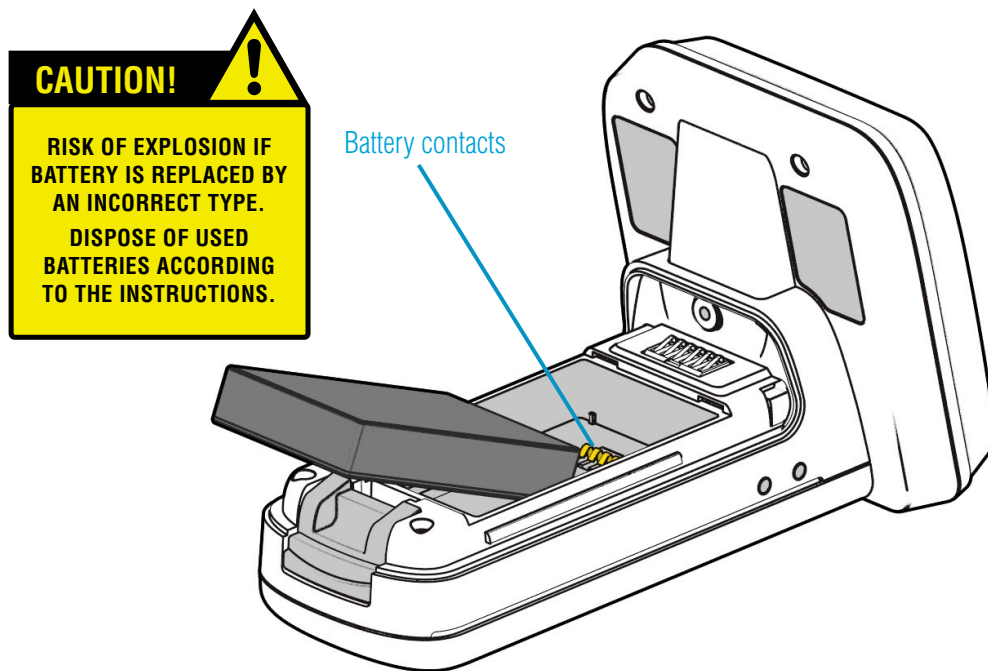


Figure 7: Battery placement

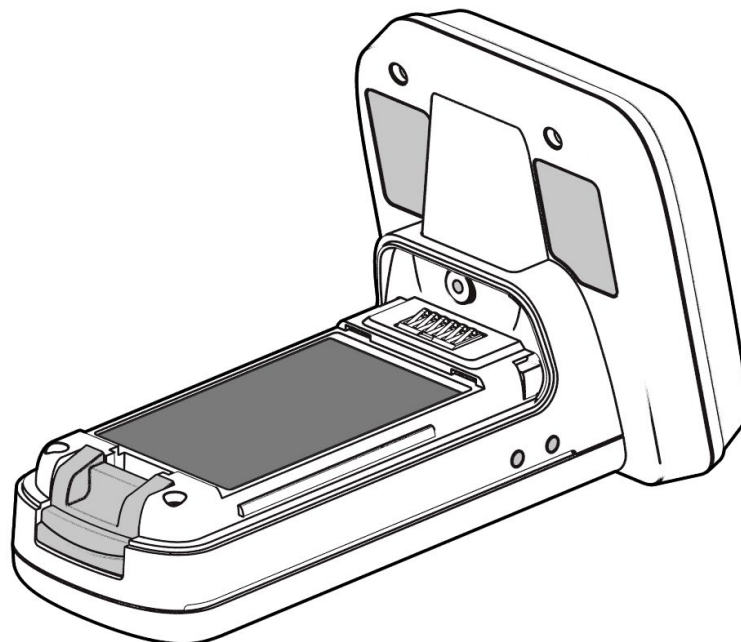


Figure 8: Battery in position

Once the battery is installed the actions to replace the cover are:

- Align the battery cover using the indents in the cover to the pips on the case
- Push the battery cover down into position
- Slide the grip handle back onto the main unit
- The release latches will audibly 'click' to indicate the grip handle is securely in place

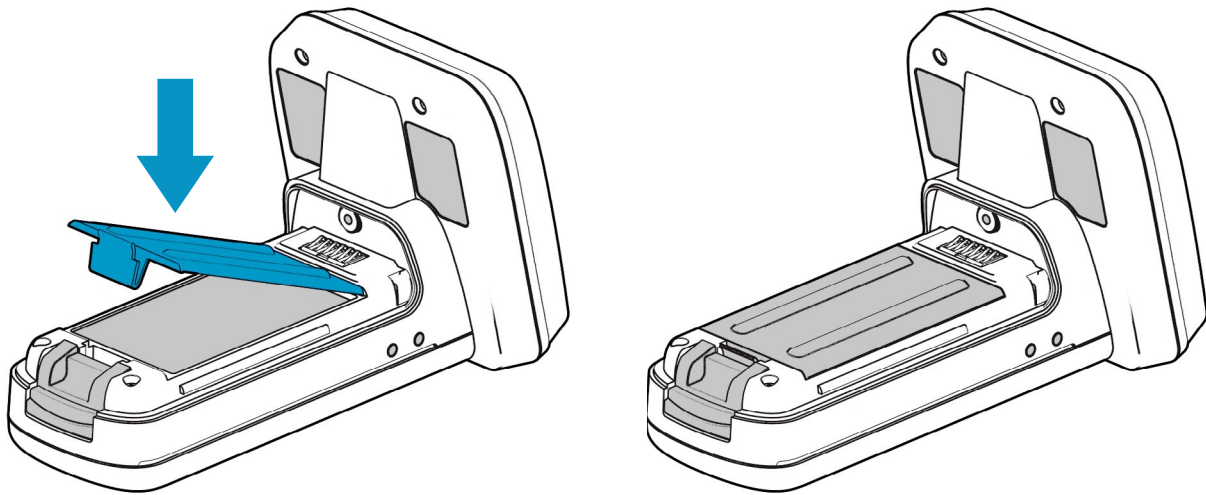


Figure 9: Replace battery cover

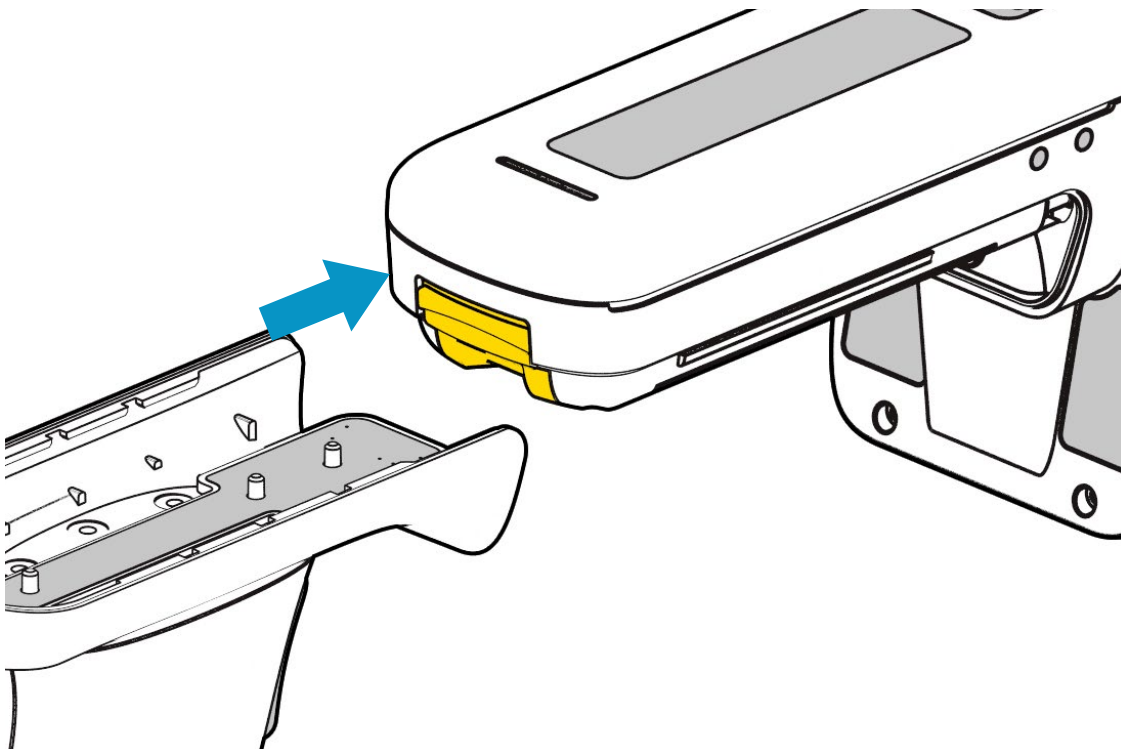


Figure 10: Slide handle into position

ANTENNA INSTALLATION OR REMOVAL

The antenna can be detached from the main body of the reader. To remove the antenna, ensure that the reader is powered off, then move the release latch and antenna in the directions indicated in the diagrams below.

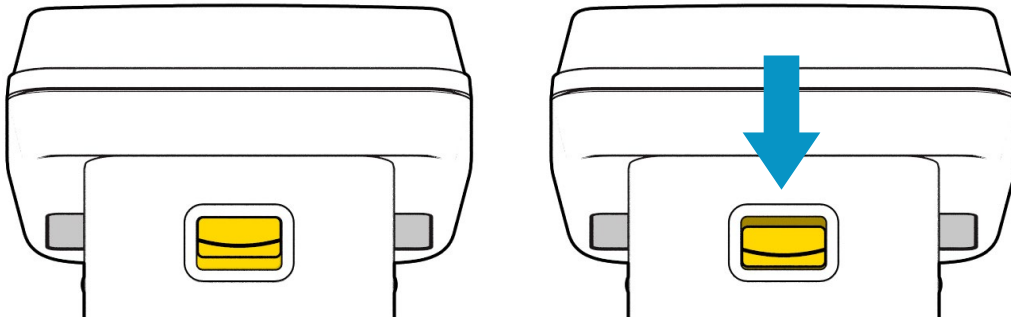


Figure 11: Antenna release latch operation

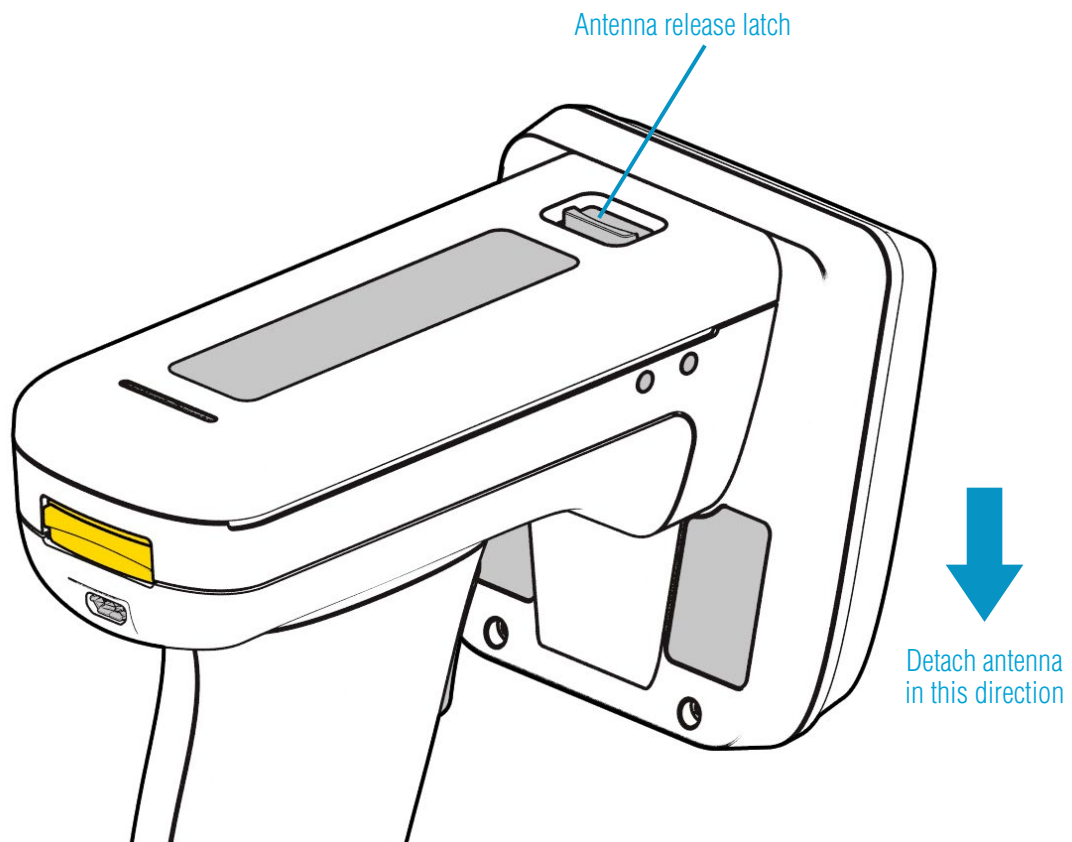


Figure 12: Antenna release latch location

LOCKING THE ANTENNA

The antenna can be locked into position using an M2.5x5mm Torx pan head screw.

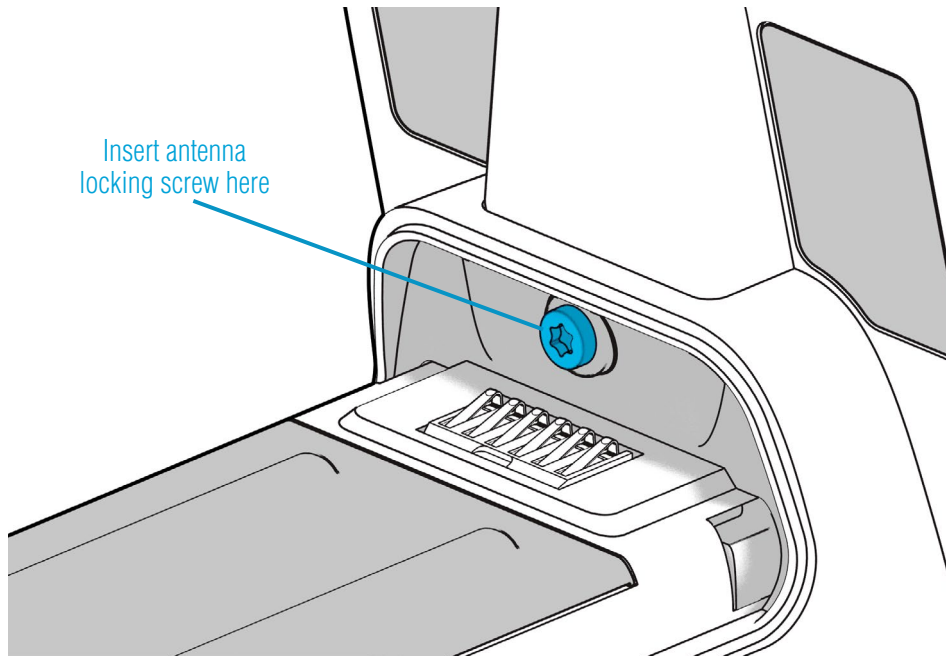
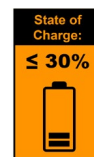


Figure 13: Antenna locking screw location

CHARGING

CONNECTING THE MICRO USB CABLE

To comply with international shipping regulations, **all batteries included with TSL products are discharged to less than 30% of their maximum capacity when shipped.** It is therefore important that the unit is fully charged before using your 1128 UHF Reader for the first time.



The 1128 UHF Reader kit is supplied with a micro USB lead for charging and synchronisation. A USB PSU is also supplied for independent charging of the 1128 UHF Reader. The Micro USB connector is inserted into the 1128 UHF Reader as shown below.

The USB Power Adaptor (PSU) should be connected to an accessible power outlet to permit easy disconnection if required.

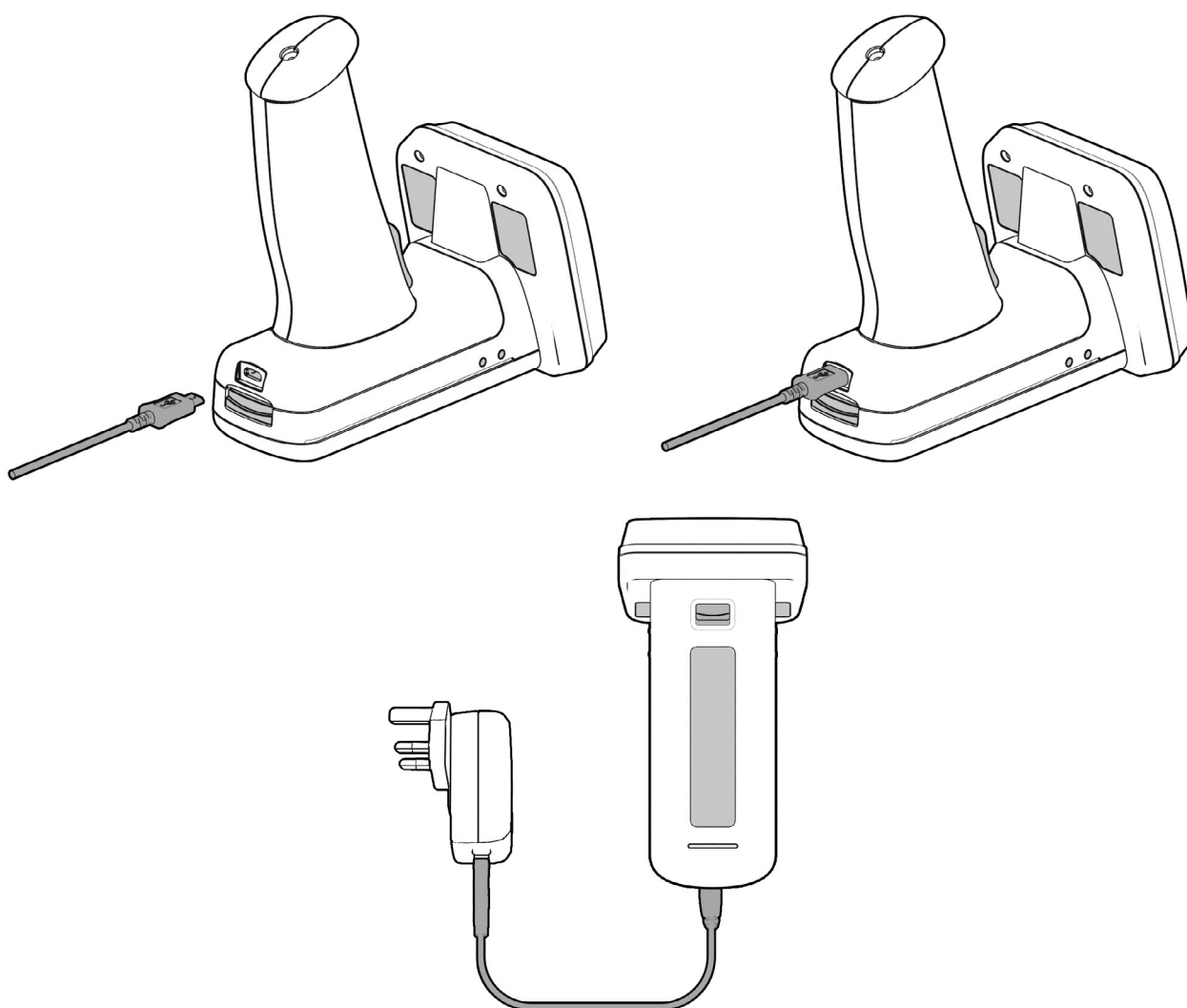
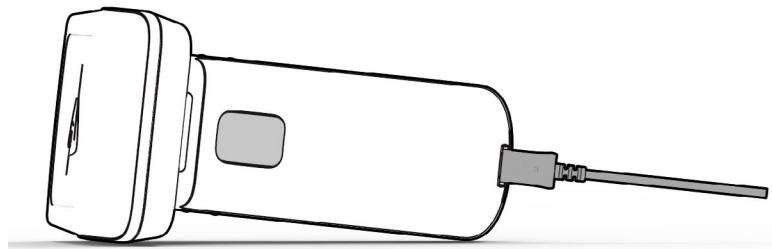


Figure 14: Attaching the Micro USB cable

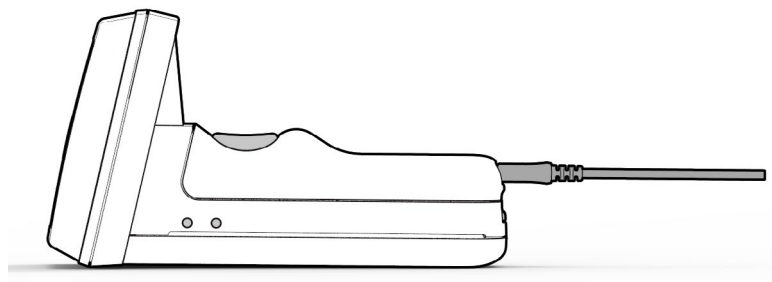
READER ORIENTATION WHEN USING MICRO USB CABLE

When charging the 1128 UHF Reader with a slimline grip, it is important to rest the device in an orientation that does not put pressure on the cable or port.



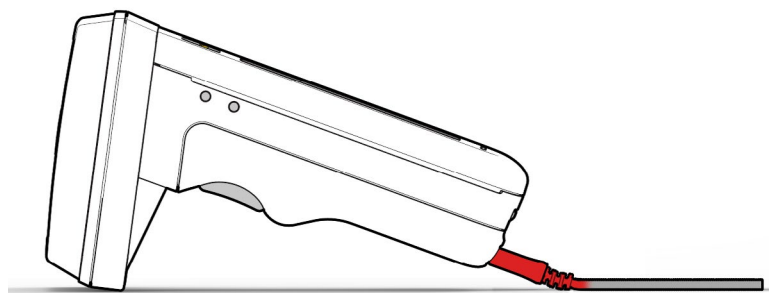
CORRECT

Device is not resting on cab



CORRECT

Device is not resting on cab



INCORRECT

Device is resting on cable causing pressure on connection.

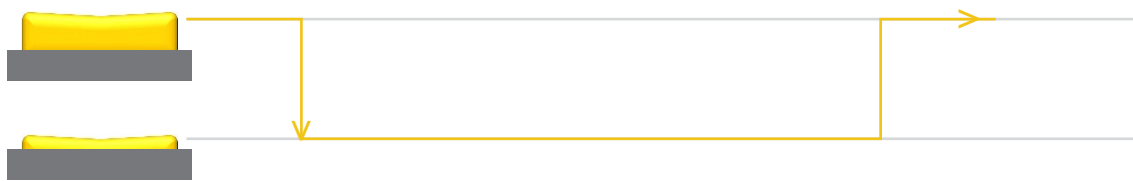
Figure 15: Micro USB cable and UHF reader orientations

BUTTON OPERATION

The 1128 UHF Reader has a Primary button action and a Secondary button action which can be initiated by different button clicks: By default, the Primary action scans for UHF transponders, whilst the Secondary action initiates the barcode scanner (Barcode scanning is only available with the 2D Imager Antenna variant). Which operation is performed depends on the way in which the button is pressed. The Single and Double press button options are also programmable.

PRIMARY BUTTON CLICK AND HOLD

- UHF Transponder Read



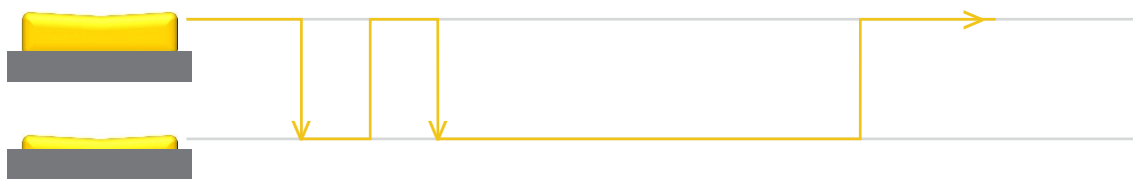
The primary button click is a standard button action:

- To initiate a primary button click press and hold the trigger button.
- To terminate a primary button click release the trigger button.

In the default configuration the 1128 UHF Reader scans for UHF transponders as the primary function. The 1128 UHF Reader will continue to scan for UHF transponders while the button is pressed. It will stop scanning once the button is released (and the current operation completes).

SECONDARY BUTTON DOUBLE CLICK AND HOLD

- Barcode Scan (2D Imager Antenna only)



The secondary button click is a single click quickly followed by a second press (press-release-press).

- To initiate a secondary button click press then release then press and hold the trigger button.
- To terminate a secondary button click release the trigger button.

In the default configuration the 1128 UHF Reader scans for barcodes as the secondary function. The 1128 UHF Reader will continue to scan for a barcode while the button is pressed. It will stop scanning when any of the following conditions are met:

- A barcode is scanned.
- The button is released.
- The barcode engine times out.

READING TRANSPONDERS

The 1128 UHF Reader can read and write to UHF RFID transponders when they are in range of the antenna. The antenna is located on the front of the 1128 UHF Reader and the read zone is in front of the antenna.

The range at which a transponder can be read depends on the transponder type and size, and the number of transponders in the field.

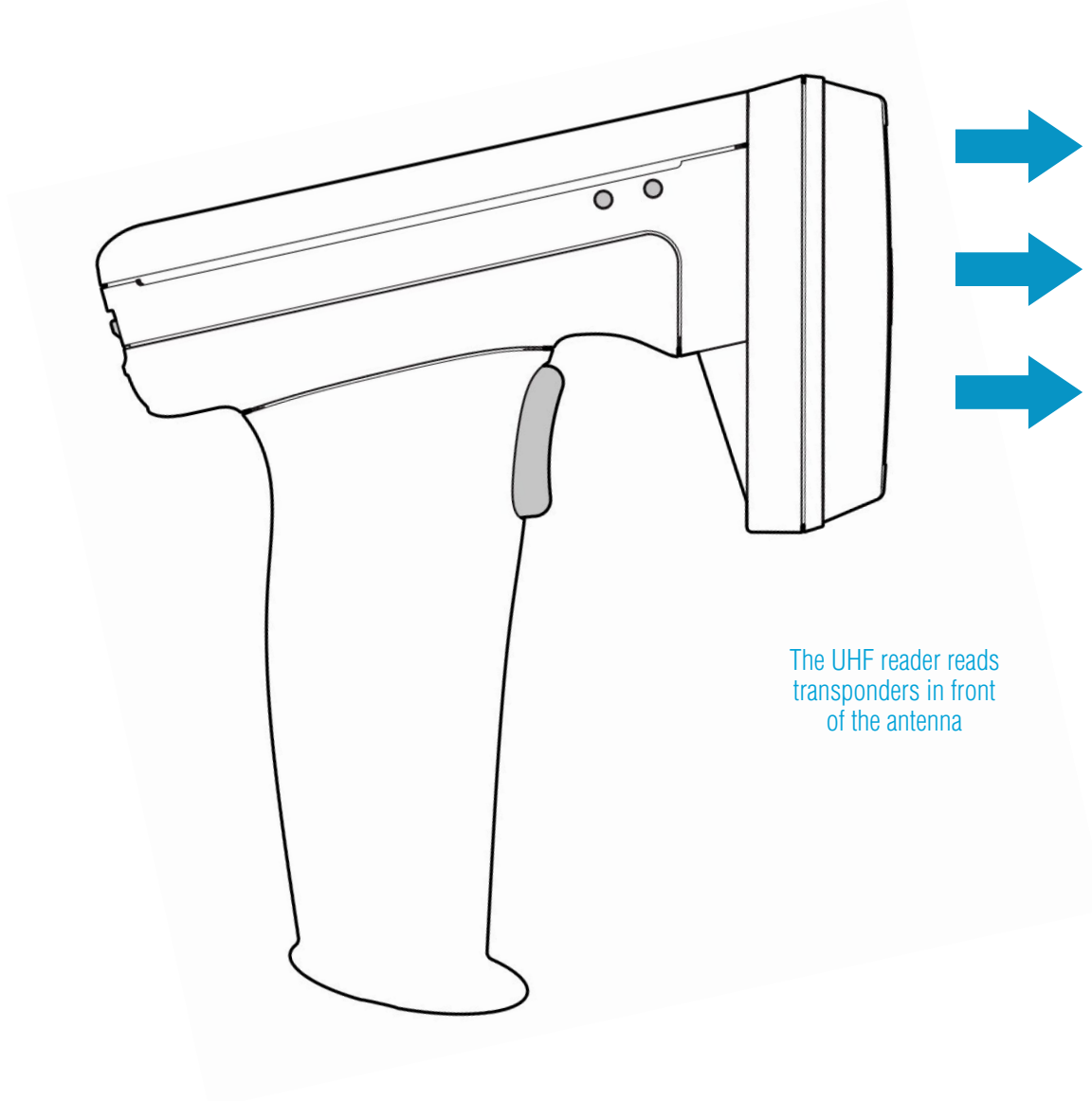


Figure 16: Antenna location and read direction

STATUS LED

The status LEDs on the top of the 1128 UHF Reader provides an indication of the operating status of the 1128 UHF Reader.

LED	Status
Blue slow flash (50% on, 50% off)	The reader is awake but there is no connection
Blue constant	The reader is awake and connected to a host
Short green flash	The reader has successfully read a tag or barcode or executed the alert command
Green slow flash (50% on, 50% off)	Antenna error
Orange slow flash (50% on, 50% off)	Battery low warning (<10% capacity remaining), please recharge immediately
Orange short single slow flash	Battery charging with battery level less than 33%
Orange short double flash	Battery charging with battery level less than or equal to 66%
Orange short triple flash	Battery charging with battery level greater than 66%
Orange rapid flash	There is a charge error / battery fault
Orange constant	The reader is fully charged
All off	The reader is off and not charging

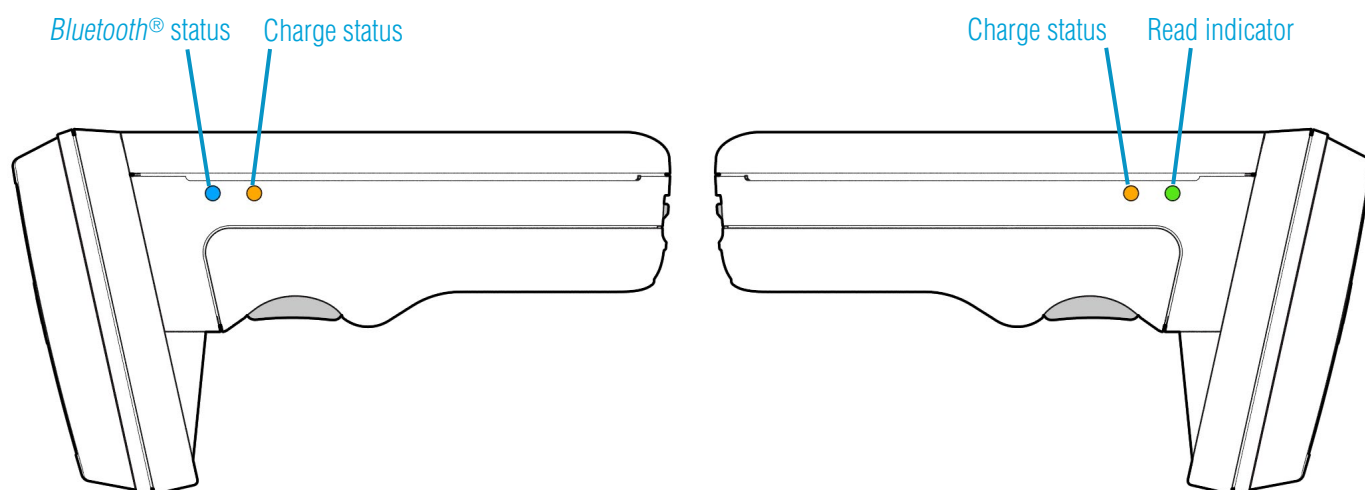


Figure 17: Location of status LEDs

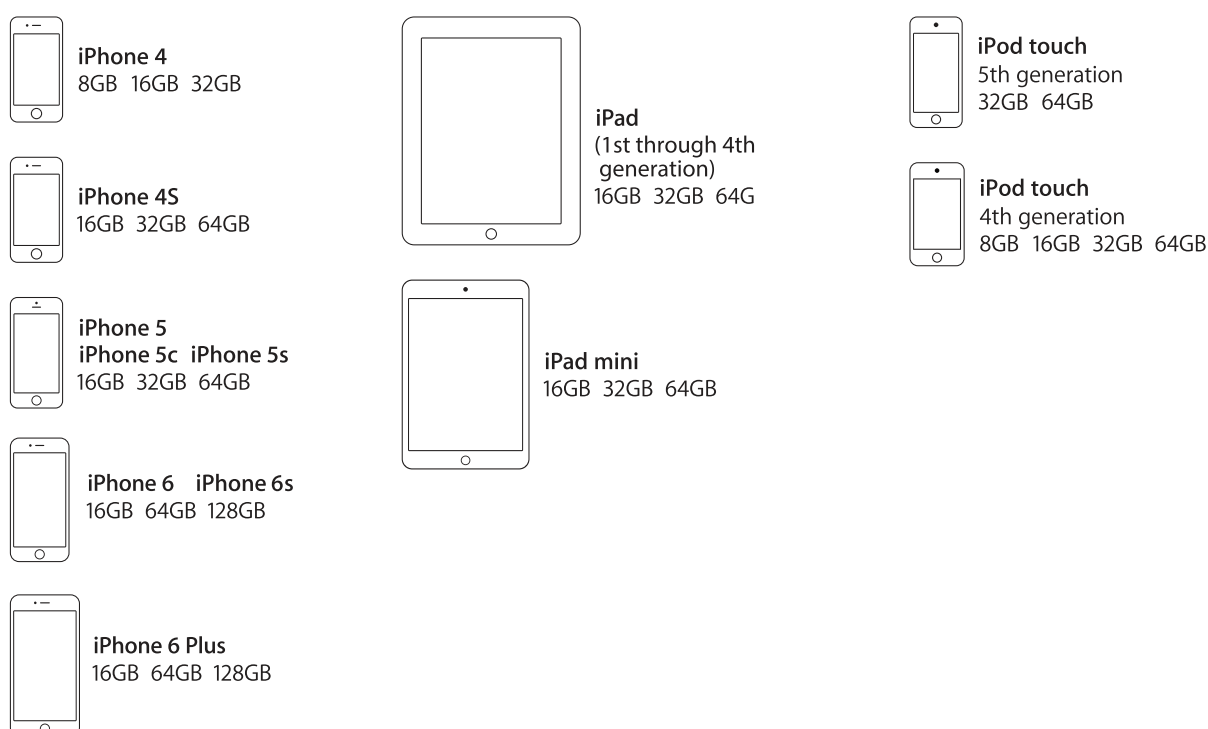
CONNECTIONS

The connection to the 1128 is either via Bluetooth or USB. Bluetooth connections use either a Virtual Com Port (VCP) or via the iAP protocol for Apple devices. Over USB only VCP is available.

For VCP the connection is established at 115200 baud, 8 data bits, no parity, 1 stop bit and RTS/CTS (hardware) flow control.

BLUETOOTH® WIRELESS TECHNOLOGY CONNECTION

COMPATIBLE APPLE PRODUCT MODELS



OTHER COMPATIBLE PRODUCTS

The 1128 *Bluetooth®* Handheld UHF RFID Reader is compatible with many other *Bluetooth®* wireless technology enabled host devices including Android, Windows CE, Windows Mobile 5/6.1/6.5 and Windows XP/Vista/7/8.

COMPARISON OF *BLUETOOTH*® OPERATING MODES

Our UHF RFID *Bluetooth*® readers support two modes of operation over *Bluetooth*®. When connected via USB the reader always supports the TSL ASCII 2 protocol, but when connected over *Bluetooth*® the reader can either be set to use the TSL ASCII 2 protocol (*Bluetooth*® SPP Mode) or it can appear as a *Bluetooth*® keyboard (*Bluetooth*® HID Mode).

The comparison chart below is designed to help you understand the differences between the two operating modes:

<i>Bluetooth</i> ® SPP Mode	<i>Bluetooth</i> ® HID Mode
Apps need to be written with specific support for the Reader	Apps can use the Reader without modification
The Reader can be discovered and paired in the Bluetooth Settings or from within the App but the App controls the connection - Connects when instructed by the App - Disconnects when instructed by the App - The reader powers off when no longer connected and idle - The App needs to connect to restore	Reader is discovered and paired in the Host Bluetooth Settings (often appears as a Keyboard) - After pairing the reader connects automatically - If idle the reader sleeps and the connection is dropped - The reader, when woken, automatically reconnects to the host device
Apps receive and interpret TSL ASCII 2 Protocol responses when the user triggers a UHF or barcode scan	Apps receive input as Key strokes from the reader including Tab and Return/Enter keys
The App can respond and react intelligently to responses e.g. - duplicate responses can be ignored or counted - incoming tag data can be truncated, stripped or transformed into alternative representations: Hex, ASCII, GS1 EPC URI, etc...	The reader types text for each barcode/UHF scan received - The tag values can be returned in Hex or ASCII representations - Up to 2 Additional characters can be inserted before and after the text sent - The reader cannot truncate or strip values from tag data
The App can change the Reader's behaviour	The Reader can only send scanned data to the host, the host cannot change the reader's behaviour
The Reader's behaviour and command parameters are controlled in real time by the App e.g. the trigger action can change to suit the task that the User is performing; the App can allow the User to specify Inventory output power	The Reader's behaviour, such as the action of the reader's trigger switch, the inventory output power, the idle sleep timeout and other command parameters can be configured only once at start-up.
Configuration is held within the App (any configuration in the AUTO.TXT is likely to be overridden by the App settings)	All configuration is held in an AUTO.TXT file stored on the (optional) SD-Card. Removing the SD-card or deleting the AUTO.TXT and power-cycling the unit restores default settings.
All reader activity, by default, is saved to a log file if an SD-Card is fitted.	All reader activity, by default, is saved to a log file if an SD-card is fitted.

CHANGING THE *BLUETOOTH*® OPERATING MODE

TSL *Bluetooth*[®] UHF RFID Readers can be operated in *SPP mode* where the Reader is controlled by a custom-written application or in *HID mode*, where the Reader behaves like a *Bluetooth*[®] Keyboard.

The HID mode has two variantsⁱ :

- | | |
|--------------------------|---|
| 1. <i>HID mode</i> | Compatible with Android, Windows and Apple devices |
| 2. <i>HID Apple mode</i> | Compatible only with Apple devices providing an on-screen keyboard toggle |

The *Bluetooth®* operating mode of a TSL UHF RFID Reader can be changed using the steps detailed below.

PREPARATION:

Download and install “TSL PC Firmware Downloader” from the Reader product download page (free, one time, registration required): www.tsl.com/1128-downloads

TO SWITCH TO *BLUETOOTH*® HID MODE:

- First delete existing pairings to the UHF reader
- Launch the Firmware Downloader (*Desktop Firmware Loader*)
- Connect the UHF reader to the computer using the supplied USB cable.
- The driver should install automatically (*)
- The downloader should prompt as the com port arrives and select the com port automatically (“COMn arrived”)
- Select **Action>Reset Bluetooth to HID mode** or **Action>Reset Bluetooth to HID Apple mode** (wait for the process to complete)
- Disconnect the UHF reader from the computer (unplug USB)
- Wake the UHF reader (pull trigger, blue flashing LED)
- Pair to the reader on the device (it typically appears as a keyboard icon and connects automatically)
- Scanned data should now be typed into the current application on the device

TO SWITCH TO *BLUETOOTH*® SPP MODE:

- First delete existing pairings to the UHF reader
- Launch the downloader (*Desktop Firmware Loader*)
- Connect the UHF reader to the computer using the supplied USB cable.
- The driver should install automatically (*)
- The downloader should prompt as the com port arrives and select the com port automatically ("COMn")

- arrived”)
- Select **Action>Reset Bluetooth to SPP mode** (wait for the process to complete)
- Disconnect the UHF reader from the computer (unplug USB)
- Wake the UHF reader (pull trigger, blue flashing LED)
- Pair to the reader on the device
- Launch an Application (e.g. *RFID Explorer*) to use the reader

(*) if not download from here: <https://ftdichip.com/drivers/vcp-drivers/>

FURTHER INFORMATION

Further information can be found from the Reader Download pages (free, one-time, registration required - see links above)

[Application Note – Bluetooth HID mode](#): Provides further examples for configuring HID mode

[Application Note – Using the Micro SD log and auto-run files](#): Describes using the AUTO.TXT

[TSL ASCII Protocol 2.5 Rev A](#): Details all the commands including for HID .st,.hs,.hc, and .hd

If you encounter any issue while following the above procedure please contact support@tsl.com providing the following information:

- The serial number of the UHF Reader
- The Firmware version of the UHF Reader
- The step at which the procedure failed
- A description of the fault that occurred

¹Firmware requirements: 1128 - v4.2.7 or greater; 1153 – v2.1.7 or greater; 1166 – v1.1.6 or greater

SETTING UP A *BLUETOOTH*® CONNECTION WITH AN APPLE PRODUCT

Awaken the 1128 UHF Reader by squeezing the trigger, which is confirmed by the flashing of the blue LED. To pair with your iOS device, navigate to the *Bluetooth*® option within the Settings menu on your iOS device.

VIEW LIST OF *BLUETOOTH*® DEVICES (IOS 7)

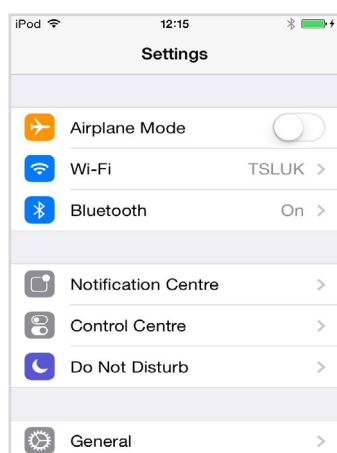
iPad® :



- Go to the Settings App.
- Select the *Bluetooth*® settings in the left hand column.

Figure 18: List of devices in *Bluetooth*® settings

iPod® / iPhone® :



- Go to the Settings App.

Figure 19: Select *Bluetooth*® settings

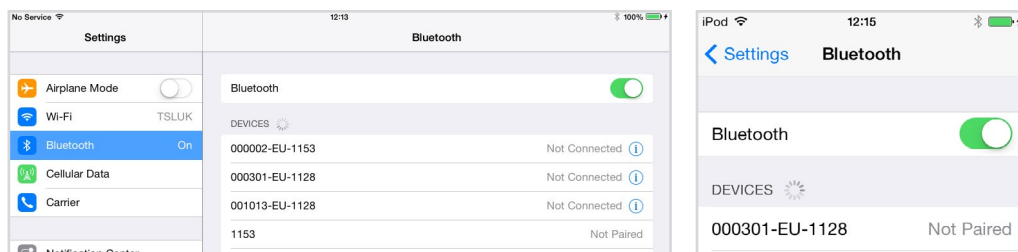


- Tap on the *Bluetooth*® row to be taken to the *Bluetooth*® settings

Figure 20: List of devices in *Bluetooth*® settings

PAIR WITH TSL *BLUETOOTH*® DEVICE

In the list of *Bluetooth*® devices, the 1128 UHF reader will be identified by its serial number (1128-xx-xxxxxx). Click on the corresponding row to pair with the reader.

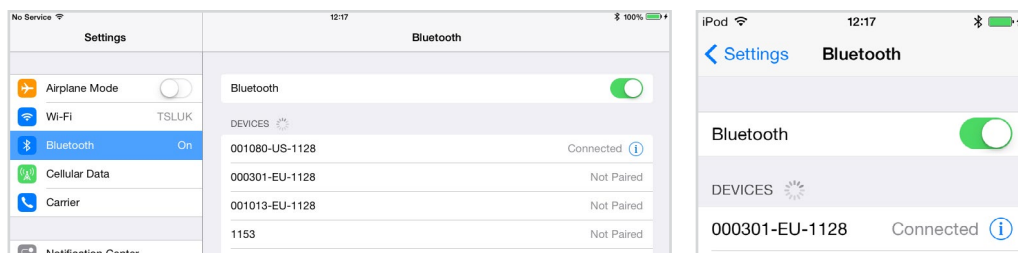


iPad®

iPod® / iPhone®

Figure 21: Identify device

After successfully pairing with the 1128 UHF Reader the device will be shown as 'connected'.

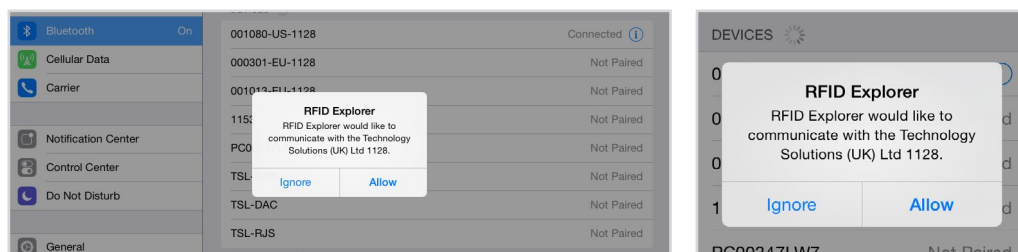


iPad®

iPod® / iPhone®

Figure 22: Device connected

If RFID Explorer is installed on your device then a prompt will appear requesting a connection to the newly added reader. Choose 'Allow' to launch RFID Explorer or 'Ignore' if the device is to be used with a different App.



iPad®

iPod® / iPhone®

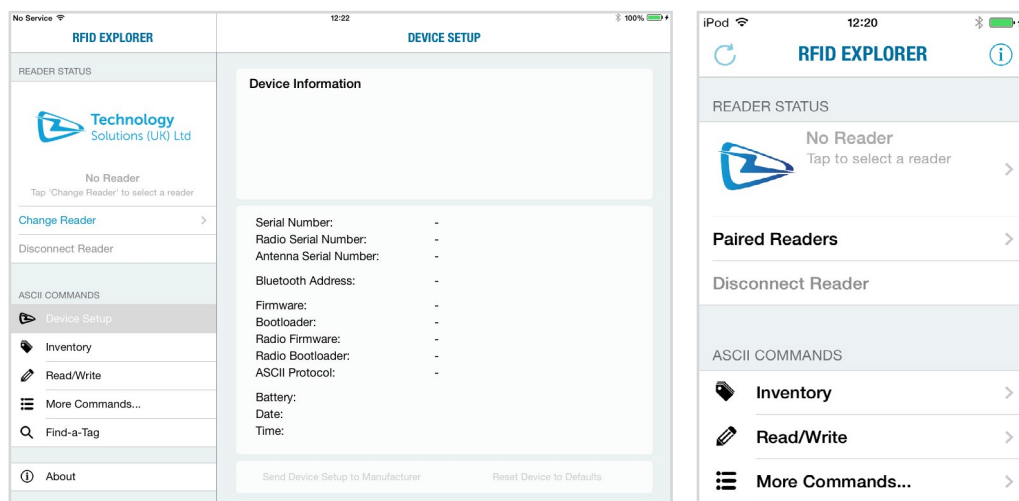
Figure 23: Allow connection



Open TSL's RFID Explorer App.

The RFID Explorer App can be downloaded from the App Store.

If the app is starting up for the first time, no reader will be selected. To select a reader, tap on 'Change Reader' in the menu to the left (iPad®) or 'Paired Readers' (iPod®/iPhone®).

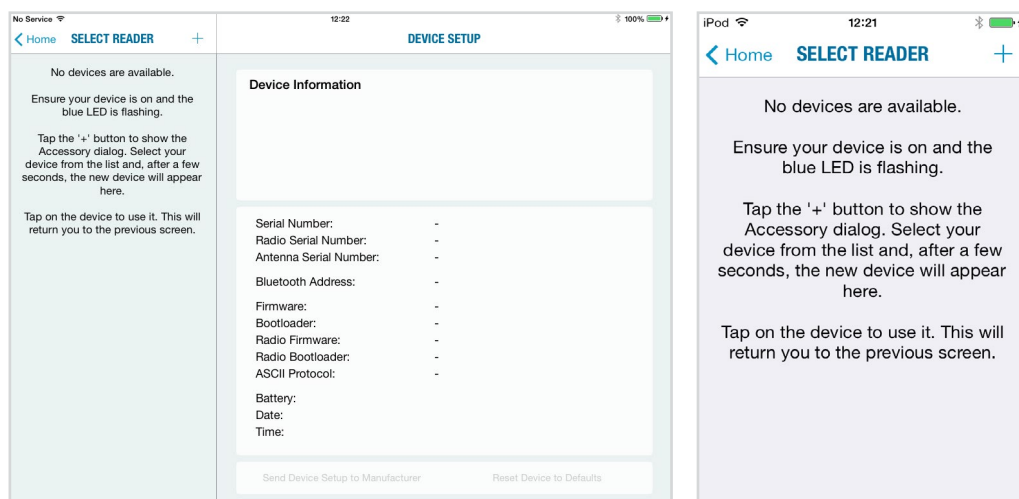


iPad®

iPod® / iPhone®

Figure 24: TSL RFID explorer App on first load - no reader selected

If you have a reader selected from your device's *Bluetooth®* settings as previously shown, continue from Figure 28. If the 'No devices are available' message in Figure 25 appears, you can also connect to a device through the RFID Explorer app. Press the + sign presented in the top right corner of the Select Reader section/screen.



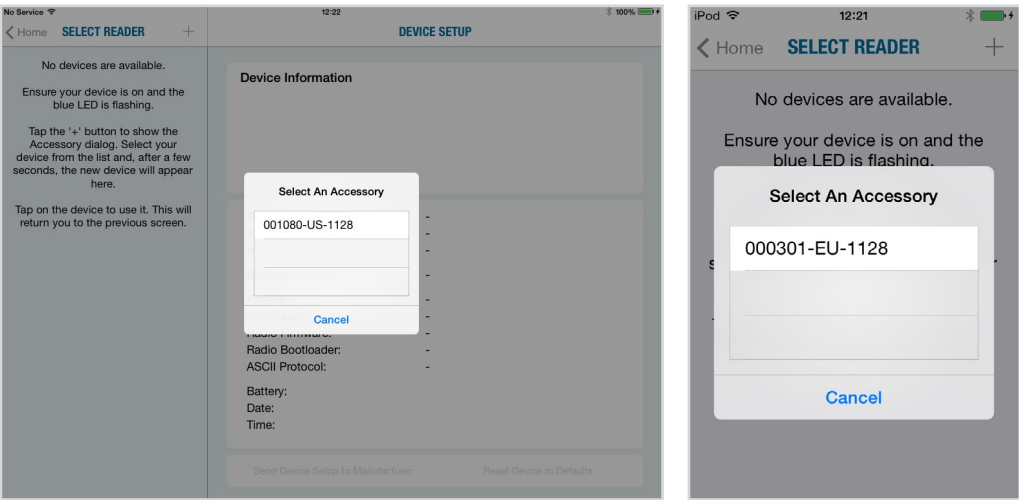
iPad®

iPod® / iPhone®

Figure 25: No devices are available

This will open a 'Select An Accessory' dialog box.
There may be a small delay in loading available readers.

Once the available readers have loaded select the 1128 UHF reader.

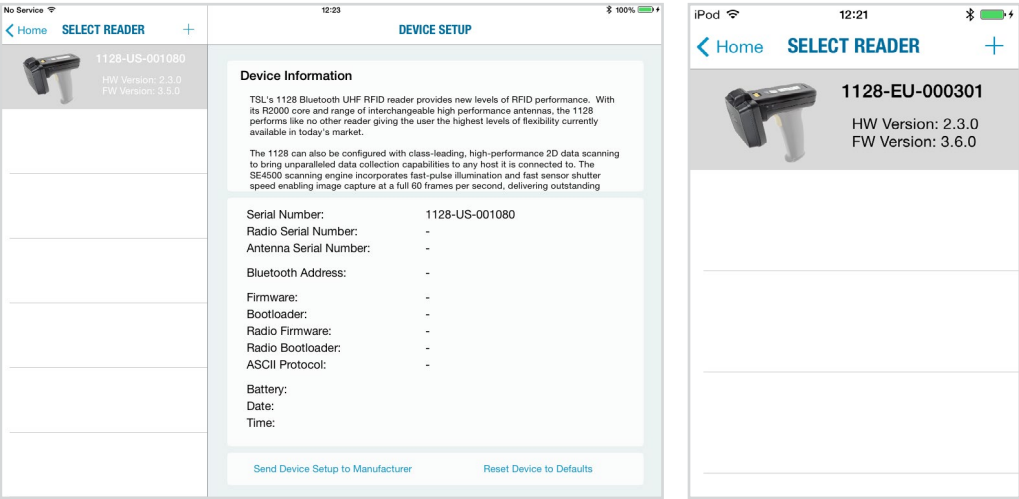


iPad®

iPod® / iPhone®

Figure 26: Select the Reader to use

A list of compatible and currently paired *Bluetooth*® devices will appear. Select the device to be used with the RFID Explorer App.

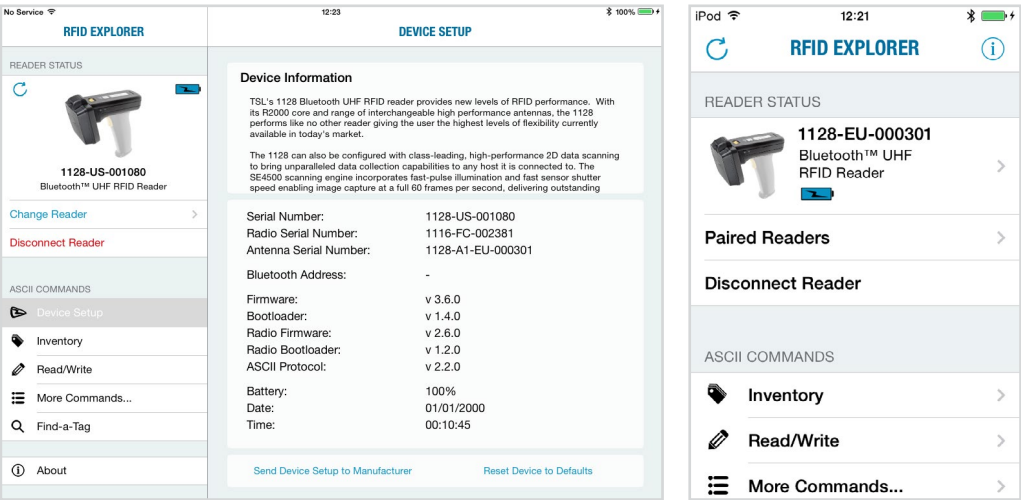


iPad®

iPod® / iPhone®

Figure 27: List of paired *Bluetooth*® devices

The selected device's image will appear, accompanied by relevant device information. The 'Inventory', 'Read/Write', 'Commands' and 'Find a Tag' features can now be explored. To maximise battery life it is recommended to release the 1128 UHF Reader from its *Bluetooth®* connection if the reader is not to be used for a significant period of time.



iPad®

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Figure 28: 1128 UHF Reader selected and ready to use

MICRO SD CARD

LOCATION, INSTALLATION AND REMOVAL

An optional Micro SD card can be installed in to the 1128 reader. The card should be SD/SDHC compatible, FAT16 or FAT32 up to 32 GB.

The card is inserted into the marked slot in the battery compartment. The Micro SD card slot is spring loaded. After inserting the card into the slot, push and release the card once to lock the card into the holder. Once locked in a further push and release will eject the card and it can then be removed from the slot.

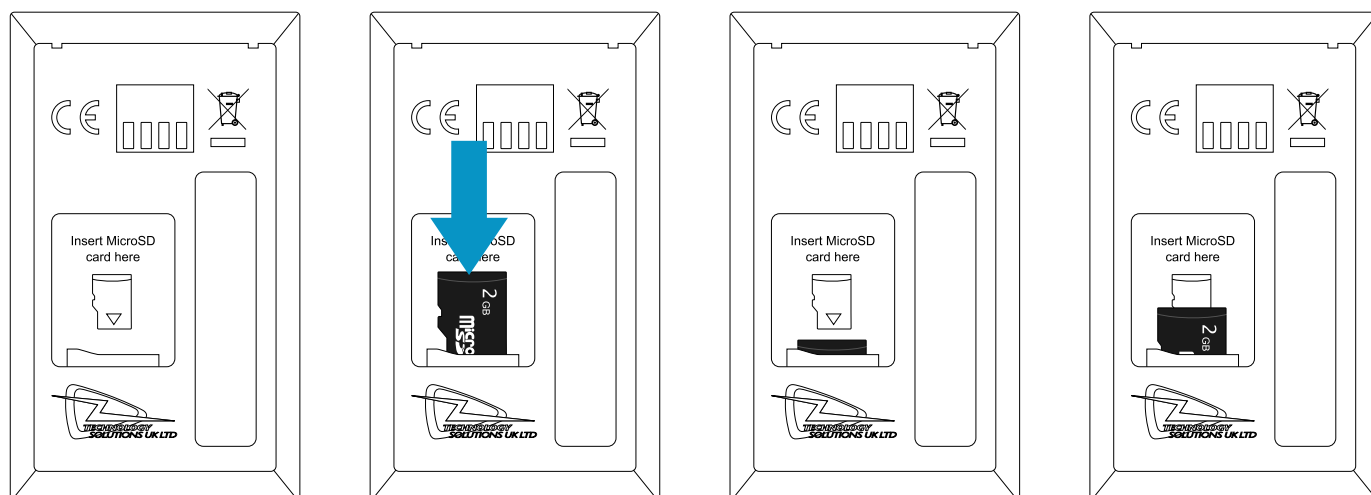


Figure 29: Micro SD Card Insertion and Removal

AUTO RUN FILE (AUTO.TXT)

Once an SD card is installed an Auto Run file can be created on the SD card. The Auto Run file contains a list of ASCII 2 commands that are executed as the 1128 powers up. The file should be located at the root of the card and called "AUTO.TXT". Although the file can contain any valid ASCII 2 commands, one per line, it is intended that these have the take no action '-n' flag specified to allow the default parameters for any command to be changed from the firmware defaults as the unit boots without actually executing the command.

For example to change the carrier power of the inventory command from 29dBm to 20dBm

'iv -o20 -n' would be appended to the Auto Run file. As is the behaviour with command parameters this will set the output power for the inventory command until a new value is sent with this command. Therefore the inventories executed by the trigger, which executes 'iv' will use the modified output power.

The Auto Run file can be manually created on the card before it is inserted into the 1128 or the ASCII 2 command Write Command to Auto Run File '.wa' can be used. The Read Auto Run File '.ra' can be used to read back the contents of the Auto Run file or delete the file from the card.

LOG FILE (LOG.TXT)

Once an SD card is installed and logging is enabled a log file is created on the SD card. The file is located at the root of the card and called "LOG.TXT". All the lines from every response to every command that is executed by the reader (including those performed from trigger actions) are appended to the end of the file. Events such as Power up are also appended to the file.

The ASCII 2 Read Log File command ('.rl') is provided to enable or disable logging, to read the log file from the card or to delete the log file from the card.

Please note that the log file can quickly become large relative to the speed at which it can be downloaded using the Read Log File Command.

Start of a sample log output:

```
DT: 2013-11-15T16:46:58
EV: Log File created
CS: .ws -sa4 -sbepc -sd300833B2DDD90140000000000 -s106 -so0002 -sts2 -dbepc -da330DE29525C0210005F5F88A
-dl06 -do0002
WW: 0
ME: No Transponder found
ER:005
DT: 2013-11-15T17:58:28
EV: Disconnected
DT: 2013-11-15T17:59:28
EV: Sleep
DT: 2013-11-18T09:18:21
EV: Wake up
CS: .al -dlon
OK:
DT: 2013-11-18T09:19:22
EV: Sleep
DT: 2013-11-18T10:07:42
EV: Wake up
DT: 2013-11-18T10:07:42
EV: Connected USB
CS: .al -dlon
OK:
CS: .iv LCMD 000000 -qss0 -qta
EP: 330DE29525C0210005F5F8F2
EP: 221001500000000000000000027
OK:
```

SOFTWARE DEVELOPMENT

To make full use of the functionality of the 1128 UHF Reader, a customised software application will be required.

The new 1128 *Bluetooth*® UHF RFID reader incorporates TSL's unique ASCII protocol for faster and easier application development. This sophisticated parameterised ASCII protocol provides the developer a powerful set of commands that carry out multiple actions locally within the *Bluetooth*® reader. This approach enables multiple tag operations executed using simple pre-configured ASCII commands which not only speeds integration of the reader into applications but also abstracts the developer from some of the complexities of the underlying Native API. Simple, text based commands are sent to the reader and responses are returned as text. This allows straightforward access to RFID tag functions such as inventory, read and write. Details of the ASCII command mode are available for download from www.tsl.com/1128-downloads.

TROUBLESHOOTING AND MAINTENANCE

MAINTENANCE

For trouble-free service please observe the following tips when using the 1128 UHF Reader:

- Protect the 1128 UHF Reader from temperature extremes. Do not leave it on the dashboard of a car on a hot day, and keep it away from heat sources.

TROUBLESHOOTING

Symptoms	Possible cause	Action
Nothing happens when the yellow button is pressed.	If the LEDs are not on then the battery may be flat.	Charge the 1128 UHF Reader.
	The 1128 UHF Reader may have button actions disabled.	Check the 1128 UHF Reader configuration and restore to defaults if unsure.
	Other.	Work through Troubleshooting Guide on TSL Website
The orange LED flashes rapidly when charging.	There is a battery fault.	Replace the battery pack.
	The battery pack temperature is outside recommended limits.	Ensure that charging only occurs between 5°C and 40°C.
The host <i>Bluetooth</i> ® discovery does not find the 1128 UHF Reader.	The 1128 UHF Reader has powered off.	Press the yellow button and ensure the blue LED is flashing.
	The 1128 UHF Reader is out of range.	Move the 1128 UHF Reader closer to the host.
	The <i>Bluetooth</i> ® friendly name of the 1128 UHF Reader has been changed.	Check the <i>Bluetooth</i> ® friendly name or restore the 1128 UHF Reader to factory defaults if unsure.
Opening the <i>Bluetooth</i> ® virtual com port does not connect to the 1128 UHF Reader.	The host has paired to a different <i>Bluetooth</i> ® device.	Pair to the required 1128 UHF Reader.
	The host <i>Bluetooth</i> ® function has an error.	Warm boot the host. If this does not help, delete the 1128 UHF Reader from the favourites list and re-pair.
<i>Bluetooth</i> ® pairing fails.	The PIN on the 1128 UHF Reader has been changed	Set the <i>Bluetooth</i> ® PIN to a known value
Other symptoms.	Other fault.	Run ASCII Remote Diagnostic and email result to support@tsl.com along with a description of the problem.

REGULATORY INFORMATION

CE MARKING AND EUROPEAN ECONOMIC AREA (EEA)

Technology Solutions UK Ltd hereby declares that this radio equipment is in compliance with Directives 2014/53/EU and 2011/65/EU.

The reader operates using the specified frequencies up to the maximum output powers as in the table below:

Radio	Frequency of operation	Maximum Output Power
Bluetooth®	2.4 GHz	3 dBm
UHF RFID	865 MHz - 868 MHz	29 dBm

INFORMATION TO THE USER – FCC

FCC warning statement:

This device complies with Part 15 of the FCC Rules.

Operation is subject to the following two conditions:

- (1) This device may not cause harmful interference, and
- (2) This device must accept any interference received, including interference that may cause undesired operation.

To comply with FCC RF exposure compliance requirements this unit must be operated in the hand with a minimum separation of 20cm from the body and other persons. Other operating configurations should be avoided. This unit must not be co-located or operated in conjunction with any other transmitter / antenna except those already approved in this filing. Handheld configurations that provide no belt-clips or other body-worn accessories and only transmit while in the hand and that maintain 20cm from the body, excluding hands, wrists, feet and ankles, are allowed.

Changes or modifications not expressly approved by the party responsible for compliance could void the user's authority to operate the equipment

RF EXPOSURE GUIDELINES

Safety information – To reduce RF exposure only use the device in accordance with the instructions supplied.

CANADIAN WARNING STATEMENTS

English

"Under Industry Canada regulations, this radio transmitter may only operate using an antenna of a type and maximum (or lesser) gain approved for the transmitter by Industry Canada.

To reduce potential radio interference to other users, the antenna type and its gain should be so chosen that the equivalent isotropically radiated power (e.i.r.p.) is not more than that

necessary for successful communication.”

“This device complies with Industry Canada licence-exempt RSS standard(s). Operation is subject to the following two conditions: (1) this device may not cause interference, and (2) this device must accept any interference, including interference that may cause undesired operation of the device.”

French

“Conformément à la réglementation d’Industrie Canada, le présent émetteur radio peut fonctionner avec une antenne d’un type et d’un gain maximal (ou inférieur) approuvé pour l’émetteur par Industrie Canada.

Dans le but de réduire les risques de brouillage radioélectrique à l’intention des autres utilisateurs, il faut choisir le type d’antenne et son gain de sorte que la puissance isotrope rayonnée équivalente (p.i.r.e.) ne dépasse pas l’intensité nécessaire à l’établissement d’une communication satisfaisante.”

“Le présent appareil est conforme aux CNR d’Industrie Canada applicables aux appareils radio exempts de licence. L’exploitation est autorisée aux deux conditions suivantes : (1) l’appareil ne doit pas produire de brouillage, et (2) l’utilisateur de l’appareil doit accepter tout brouillage radioélectrique subi, même si le brouillage est susceptible d’en compromettre le fonctionnement.”

KOREA WARNING STATEMENT FOR CLASS B ITE

기종별	사용자안내문
B 급 기기 (가정용 방송통신기기)	이 기기는 가정용 (B 급) 으로 전자파적합등록을 한 기기로서 주로 가정에서 사용하는 것을 목적 으로 하며 , 모든 지역에서 사용할 수 있습니다 .

BATTERY SAFETY



- Disposal of a battery into fire or a hot oven, or mechanically crushing or cutting a battery, can result in an explosion.
- Leaving a battery in an extremely high temperature environment can result in an explosion or the leakage of flammable liquid or gas.
- A battery subjected to extremely low air pressure may result in an explosion or the leakage of flammable liquid or gas.

HEALTH AND SAFETY RECOMMENDATIONS

Ergonomic Recommendations

Caution: In order to avoid or minimize the potential risk of ergonomic injury, follow the recommendations below. Consult with your local Health & Safety Manager to ensure that you are adhering to your company's safety programs to prevent employee injury.

- Reduce or eliminate repetitive motion
- Maintain a natural position
- Reduce or eliminate excessive force
- Keep objects that are used frequently within easy reach Perform tasks at correct heights
- Reduce or eliminate vibration
- Reduce or eliminate direct pressure
- Provide adjustable workstations
- Provide adequate clearance
- Provide a suitable working environment
- Improve work procedures.

For vehicle installation and use

An air bag inflates with great force. DO NOT place objects, including either installed or portable wireless equipment, in the area over the air bag or in the air bag deployment area. If in-vehicle wireless equipment is improperly installed and the air bag inflates, serious injury could result.

RF signals may affect improperly installed or inadequately shielded electronic systems in motor vehicles (including safety systems). Check with the manufacturer or its representative regarding your vehicle. You should also consult the manufacturer of any equipment that has been added to your vehicle.

Power Supply

Use only TSL-approved cradles, chargers and power supplies with the 1128 UHF Reader. Use of an alternative power supply will invalidate any approval given to this device, void the warranty for the product and may be dangerous.

WASTE ELECTRICAL AND ELECTRONIC EQUIPMENT (WEEE)

For EU Customers: All products at the end of their life must be returned to TSL for recycling. For information on how to return product please contact TSL.

WARRANTY

(A) Warranty

TSL's hardware Products are warranted against defects in workmanship and materials for a period of twelve (12) months from the date of shipment, unless otherwise provided by TSL in writing, provided the Product remains unmodified and is operated under normal and proper conditions. Warranty provisions and durations on software, integrated installed systems, Product modified or designed to meet specific customer specifications ("Custom Products"), remanufactured products, and reconditioned or upgraded products, shall be as provided in the applicable Product specification in effect at the time of purchase or in the accompanying software license.

(B) Spare Parts

Spare parts (i.e. parts, components, or subassemblies sold by TSL for use in the service and maintenance of Products) are warranted against defects in workmanship and materials for a period of thirty (30) days from the date of shipment. Spare parts may be new or originate from returned units under the conditions set forth in subsection D below.

(C) Repair of TSL branded hardware

For repairs on TSL branded hardware Products under this Agreement, including repairs covered by warranty, the repair services provided are warranted against defects in workmanship and materials on the repaired component of the Product for a period of thirty (30) days from the shipment date of the repaired Product, or until the end of the original warranty period, whichever is longer. Any such defects shall be notified to TSL in writing within 7 days of the same becoming apparent.

(D) Product Service

Products may be serviced or manufactured with parts, components, or subassemblies that originate from returned products and that have been tested as meeting applicable specifications for equivalent new material and Products. The sole obligation of TSL for defective hardware Products is limited to repair or replacement (at TSL's option) on a "return to base (RTB)" basis with prior TSL authorisation.

Customer is responsible for prompt shipment to TSL and assumes all costs and risks associated with this transportation; return shipment to the Customer will be at TSL's expense. Customer shall be responsible for return shipment charges for product returned where TSL determines there is no defect ("No Defect Found"), or for product returned that TSL determines is not eligible for warranty repair. No charge will be made to Buyer for replacement parts for warranty repairs. TSL is not responsible for any damage to or loss of any software programs, data or removable data storage media, or the restoration or reinstallation of any software programs or data other than the software, if any, installed by TSL during manufacture of the Product.

(E) Original Warranty Period

Except for the warranty applying solely to the repaired component arising from a repair service as provided in Section C above, the aforementioned provisions do not extend the original warranty period of any Product that had either been repaired or replaced by TSL.

(F) Warranty Provisions

The above warranty provisions shall not apply to any Product:

- (i) which has been repaired, tampered with, altered or modified, except by TSL's authorized service personnel;
- (ii) in which the defects or damage to the Product result from normal wear and tear, misuse, negligence, improper storage, water or other liquids, battery leakage, use of parts or accessories not approved or supplied by TSL, or failure to perform operator handling and scheduled maintenance instructions supplied by TSL;
- (iii) which has been subjected to unusual physical or electrical stress, abuse, or accident, or forces or exposure beyond normal use within the specified operational and environmental parameters set forth in the applicable Product specification; nor shall the above warranty provisions apply to any expendable or consumable items, such as batteries, supplied with the Product.

EXCEPT FOR THE WARRANTY OF TITLE AND THE EXPRESS WARRANTIES STATED ABOVE, TSL DISCLAIMS ALL WARRANTIES ON PRODUCTS FURNISHED HEREUNDER INCLUDING ALL IMPLIED WARRANTIES OF MERCHANTABILITY AND FITNESS FOR A PARTICULAR USE. ANY IMPLIED WARRANTIES THAT MAY BE IMPOSED BY LAW ARE LIMITED IN DURATION TO THE LIMITED WARRANTY PERIOD. SOME STATES OR COUNTRIES DO NOT ALLOW A LIMITATION ON HOW LONG AN IMPLIED WARRANTY LASTS OR THE EXCLUSION OR LIMITATION OF INCIDENTAL OR CONSEQUENTIAL DAMAGES OR CONSUMER PRODUCTS. IN SUCH STATES OR COUNTRIES, FOR SUCH PRODUCTS, SOME EXCLUSIONS OR LIMITATIONS OF THIS LIMITED WARRANTY MAY NOT APPLY.

The stated express warranties are in lieu of all obligations or liabilities on the part of TSL for damages, including but not limited to, special, indirect or consequential damages arising out of or in connection with the use or performance of the Product or service. TSL's liability for damages to Buyer or others resulting from the use of any Product or service furnished hereunder shall in no way exceed the purchase price of said Product or the fair market value of said service, except in instances of injury to persons or property.

TSL is not responsible for any damages incurred during shipment if the approved shipping container is not used. Shipping the units improperly can possibly void the warranty. If the original shipping container was not kept, contact your local distributor or TSL to have another sent to you.

TSL shall not be responsible for any injury, damage or loss of whatever kind caused directly or indirectly by the goods whether as a result of their manufacture, operation, use or otherwise and the customer shall indemnify TSL from any claim arising from any loss suffered by any third party.

ABOUT

ABOUT TSL



Technology Solutions UK Ltd (TSL), part of HID, is a leading manufacturer of high performance mobile RFID readers used to identify and track products, assets, data or personnel.

For over two decades, TSL has delivered innovative data capture solutions to Fortune 500 companies around the world using a global network of distributors and system integrators. Specialist in-house teams design all aspects of the finished products and software ecosystems, including electronics, firmware, application development tools, RF design and injection mould tooling.

TSL is an ISO 9001:2015 certified company.



ISO 9001: 2015

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ABOUT HID



HID powers the trusted identities of the world's people, places and things. We make it possible for people to transact safely, work productively and travel freely. Our trusted identity solutions give **people** convenient access to physical and digital **places** and connect **things** that can be identified, verified and tracked digitally. Millions of people around the world use HID products and services to navigate their everyday lives, and billions of things are connected through HID technology. We work with governments, educational institutions, hospitals, financial institutions, industrial businesses and some of the most innovative companies on the planet. Headquartered in Austin, Texas, HID has over 4,000 employees worldwide and operates international offices that support more than 100 countries. HID is an ASSA ABLOY Group brand.

For more information, visit www.hidglobal.com.

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